

MATH I – Week 4

Fall 2026

NOTES

INTRODUCTION: This set of Notes has been prepared by G. Neofotistos to a) summarize the general properties of limits (summarize the fundamental concept), b) present important standard limits, c) present and discuss indeterminate forms, asymptotics and asymptotic equivalence, and d) introduce the Landau symbols notation.

General properties of limits (a summary of the fundamental concepts)

Limits describe the value that a function approaches when its independent variable approaches a specified point. If

$$\lim_{x \rightarrow a} f(x) = L,$$

then $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a , without requiring $x = a$. When a limit exists, it is unique. A two-sided limit exists only when the left- and right-hand limits exist and are equal:

$$\lim_{x \rightarrow a} f(x) = L \Leftrightarrow \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

Limits obey algebraic rules. If

$$\lim_{x \rightarrow a} f(x) = L, \lim_{x \rightarrow a} g(x) = M,$$

then

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) + g(x)] &= L + M, \\ \lim_{x \rightarrow a} f(x)g(x) &= LM, \end{aligned}$$

and, provided $M \neq 0$,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Similar rules apply to differences, constant multiples, powers and roots. For functions that are continuous at a , the limit can usually be calculated by direct substitution:

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Standard limits

Standard limits are fundamental results that can be used to evaluate more complicated limits. Some of the most important are

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1,$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2},$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1,$$

$$\lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = 1,$$

and

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

For the trigonometric limits, angles must be measured in radians.

These standard limits can be adapted by substitution to produce results fast and also develop intuition. For example,

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x} = 5 \lim_{x \rightarrow 0} \frac{\sin(5x)}{5x} = 5.$$

Another important family concerns exponential growth:

$$\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0$$

for every positive integer n . This leads to comprehend that exponential functions eventually grow faster than any fixed polynomial.

Indeterminate forms

An **indeterminate form** occurs when direct substitution does not determine the value of a limit, because it can lead to indeterminate forms of the type:

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty - \infty, 0^0, 1^\infty, \infty^0.$$

The word *indeterminate* does not mean that the limit does not exist. It means that more analysis is required. For example,

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

has the form $0/0$. *Factoring* gives

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1$$

for $x \neq 1$. Therefore,

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

However,

$$\lim_{x \rightarrow 0} \frac{x}{x} = 1, \lim_{x \rightarrow 0} \frac{x^2}{x} = 0, \lim_{x \rightarrow 0} \frac{x}{x^2}$$

does not exist as a finite two-sided limit, although all three expressions initially produce $0/0$. This demonstrates why $0/0$ cannot be assigned a fixed value.

Depending on the expression, indeterminate forms may be resolved through *factoring*, *rationalization*, *standard limits*, *asymptotic equivalence* (which will be discussed below), *changes of variable*, *algebraic rearrangement*, *the Squeeze Theorem*, and Taylor expansions or, when its hypotheses are satisfied, L'Hôpital's rule (but the latter ones will be discussed later in the Math I course).

About asymptotics

Asymptotic analysis **compares functions near a particular point**—often as

$$x \rightarrow 0, x \rightarrow a, \text{ or } x \rightarrow \infty.$$

Instead of asking whether two functions are exactly equal, we ask **whether they have approximately the same behaviour in the limit**.

For example, when x is close to zero,

$$\sin x \approx x.$$

This can be expressed precisely using asymptotic equivalence.

Asymptotic equivalence

We say that $f(x)$ and $g(x)$ are **asymptotically equivalent** as $x \rightarrow a$, and write

$$f(x) \sim g(x) \quad (\text{as } x \rightarrow a),$$

if

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 1.$$

This means that the relative difference between the functions approaches zero.

Equivalently,

$$f(x) = g(x)(1 + \varepsilon(x)),$$

where

$$\varepsilon(x) \rightarrow 0.$$

See also (reference: https://en.wikipedia.org/wiki/Asymptotic_analysis)

Definition [edit]

Formally, given functions $f(x)$ and $g(x)$, we define a [binary relation](#)

$$f(x) \sim g(x) \quad (\text{as } x \rightarrow \infty)$$

if and only if (de Bruijn 1981, §1.4)

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1.$$

The symbol $\sim$ is the [tilde](#). The relation is an [equivalence relation](#) on the set of functions of x ; the functions f and g are said to be *asymptotically equivalent*. The [domain](#) of f and g can be any set for which the limit is defined: e.g. real numbers, complex numbers, positive integers.

The same notation is also used for other ways of passing to a limit: e.g. $x \rightarrow 0$, $x \downarrow 0$, $|x| \rightarrow 0$. The way of passing to the limit is often not stated explicitly, if it is clear from the context.

Although the above definition is common in the literature, it is problematic if $g(x)$ is zero infinitely often as x goes to the limiting value. For that reason, some authors use an alternative definition. The alternative definition, in [little-o notation](#), is that $f \sim g$ if and only if

$$f(x) = g(x)(1 + o(1)).$$

This definition is equivalent to the prior definition if $g(x)$ is not zero in some [neighbourhood](#) of the limiting value.^{[5]:§1.2}

We are going to discuss little-o notation later (this document).

Example: $\sin x$ and x

The standard limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

means that

$$\boxed{\sin x \sim x \quad (\text{as } x \rightarrow 0).}$$

The statement does not mean that $\sin x = x$. It means that their ratio approaches 1 near zero.

For example,

$$\frac{\sin(0.01)}{0.01} \approx 0.999983.$$

Standard asymptotic equivalences

Remember the standard limits...

As $x \rightarrow 0$,

$$\sin x \sim x,$$

$$\tan x \sim x,$$

$$1 - \cos x \sim \frac{x^2}{2},$$

$$e^x - 1 \sim x,$$

$$\ln(1 + x) \sim x,$$

and, for a constant α ,

$$(1 + x)^\alpha - 1 \sim \alpha x.$$

The trigonometric equivalences require angles to be measured in radians.

Using asymptotic equivalence to calculate limits

Consider

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{5x}.$$

Since

$$\sin(3x) \sim 3x,$$

we obtain

$$\frac{\sin(3x)}{5x} \sim \frac{3x}{5x} = \frac{3}{5}.$$

Therefore,

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{5x} = \frac{3}{5}.$$

Another example is

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}.$$

Using

$$1 - \cos x \sim \frac{x^2}{2},$$

we get

$$\frac{1 - \cos x}{x^2} \sim \frac{x^2/2}{x^2} = \frac{1}{2}.$$

Thus,

$$\boxed{\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

5. Equivalence at infinity

Asymptotic equivalence is also useful when $x \rightarrow \infty$.

For example,

$$3x^2 + 5x - 1 \sim 3x^2 \ (x \rightarrow \infty),$$

because

$$\frac{3x^2 + 5x - 1}{3x^2} = 1 + \frac{5}{3x} - \frac{1}{3x^2} \rightarrow 1.$$

Thus, the highest-degree term controls the long-term behaviour of a polynomial.

Similarly,

$$\frac{2x^3 + x}{5x^2 - 1} \sim \frac{2x^3}{5x^2} = \frac{2}{5}x.$$

Landau symbols

Landau symbols describe the relative size or growth rate of functions. The two most important symbols in introductory calculus are:

$$O(g(x)) \text{ and } o(g(x)).$$

Big- O notation

We write

$$\boxed{f(x) = O(g(x)) \ (x \rightarrow a)}$$

if f is eventually no larger than a constant multiple of g , in absolute value.

Informally,

$$f(x) = O(g(x))$$

means: **f has at most the order of magnitude of g .**

Example near zero

Since

$$|\sin x| \leq |x|,$$

we can write

$$\sin x = O(x) \quad (x \rightarrow 0).$$

Example at infinity

As $x \rightarrow \infty$,

$$3x^2 + 5x - 1 = O(x^2).$$

The lower-degree terms do not cause the polynomial to grow faster than a constant multiple of x^2 .

Little-*o* notation

We write

$$f(x) = o(g(x)) \quad (\text{as } x \rightarrow a)$$

if

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0.$$

This means that $f(x)$ becomes negligible relative to $g(x)$.

Example near zero

As $x \rightarrow 0$,

$$x^2 = o(x),$$

because

$$\frac{x^2}{x} = x \rightarrow 0.$$

Similarly,

$$x^3 = o(x^2)$$

because

$$\frac{x^3}{x^2} = x \rightarrow 0.$$

Example at infinity

As $x \rightarrow \infty$,

$$x = o(x^2),$$

because

$$\frac{x}{x^2} = \frac{1}{x} \rightarrow 0.$$

Also,

$$x^{10} = o(e^x) \quad (\text{as } x \rightarrow \infty),$$

because exponential growth eventually dominates every fixed power of x .

8. Difference between O and o

The distinction is important:

$$f(x) = O(g(x))$$

means that f is bounded by a constant multiple of g , whereas

$$f(x) = o(g(x))$$

means that f becomes negligible compared with g .

For example, as $x \rightarrow 0$,

$$3x = O(x),$$

but

$$3x \neq o(x),$$

because

$$\frac{3x}{x} = 3,$$

which does not approach zero.

However,

$$x^2 = o(x).$$

Every little- o relation implies the corresponding big- O relation:

$$f = o(g) \Rightarrow f = O(g),$$

but the converse is not generally true.

Relationship with asymptotic equivalence

Asymptotic equivalence can be written using little- o notation:

$$f(x) \sim g(x)$$

if and only if

$$f(x) = g(x) + o(g(x)).$$

For example,

$$\sin x \sim x$$

can be written as

$$\sin x = x + o(x) \quad (as \ x \rightarrow 0).$$

Similarly,

$$e^x = 1 + x + o(x) \quad (x \rightarrow 0),$$

because

$$e^x - 1 - x$$

is negligible compared with x .

A more accurate approximation is

$$e^x = 1 + x + \frac{x^2}{2} + o(x^2).$$

Important caution

The notation

$$f(x) = O(g(x))$$

does not represent an ordinary equation. The symbol $O(g(x))$ describes an entire class of functions.

For example,

$$2x^2 + 3x + 1 = 2x^2 + O(x) \quad (as \ x \rightarrow \infty)$$

means that the remaining part $3x + 1$ grows no faster than a constant multiple of x .

A useful summary is:

$$f \sim g: \frac{f}{g} \rightarrow 1$$

$$f = o(g): \frac{f}{g} \rightarrow 0$$

$$f = O(g): \left| \frac{f}{g} \right| \text{ remains bounded.}$$

Also (in conversation with Gemini):

Landau symbols (asymptotic notation) describe how functions or algorithms behave as inputs grow very large or approach a limit.

Main Landau Symbols in Mathematics and Science

In mathematics and physics, Landau symbols describe the asymptotic behavior of functions (typically as $x \rightarrow \infty$ or $x \rightarrow 0$). Let $f(x)$ and $g(x)$ be two functions.

Symbol	Name	Mathematical Meaning	Informal Description
$f(x) = O(g(x))$	Big-O	$ f(x) \leq M g(x) $ for some constant M as $x \rightarrow x_0$.	f grows at most as fast as g (upper bound).
$f(x) = o(g(x))$	Small-o	$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0$.	f grows strictly slower than g (negligible compared to g).
$f(x) \sim g(x)$	Asymptotic equivalence	$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1$.	f and g approach the exact same growth rate.

Regarding use of Landau symbols in computer science, see from OpenStax “Introduction to Computer Science” Section 3.3 — Formal Properties of Algorithms:

Another way to understand orders of growth is to consider how a change in the size of the problem results in a change to the resource usage. When we double the size of the input problem, algorithms in each order of growth respond differently ([Figure 3.15](#)).

- $O(1)$ algorithms will not require any more resources.
- $O(\log N)$ algorithms will require 1 additional resource unit.
- $O(N)$ algorithms will require 2 times the number of resources.
- $O(N \log N)$ algorithms will require a little more than 2 times the number of resources.
- $O(N^2)$ algorithms will require 4 times the number of resources.
- $O(N^3)$ algorithms will require 8 times the number of resources.
- $O(2^N)$, $O(3^N)$, . . . algorithms will require the squared or cubed number of resources.
- $O(N!)$ algorithms will require even more.

Order of growth	Algorithm	Execution time for $N = 10$	Execution time for $N = 1,000,000$
Constant	$O(1)$	1 μ sec	1 μ sec
Logarithmic	$O(\log N)$	3 μ sec	18 μ sec
Linear	$O(N)$	10 μ sec	1 sec
Log-linear	$O(N \log N)$	33 μ sec	19.8 sec
Quadratic	$O(N^2)$	100 μ sec	11.6 days
Cubic	$O(N^3)$	1 msec	31.7 years
Exponential	$O(2^N)$	10 msec	Billion of years
Factorial	$O(N!)$	3.6 sec	Practically infinite

Figure 3.15 This chart shows the time it would take for an algorithm with each of the given orders of growth to finish running on a problem of the given size, N . When an algorithm takes longer than 1025 years to compute, that means it takes longer than the current age of the universe. (data source: Geeks for Geeks, "Big O Notation Tutorial—A Guide to Big O Analysis," last updated March 29, 2024; attribution: Copyright Rice University, OpenStax, under CC BY NC-SA 4.0 license)

This set of Notes has been produced by G. Neofotistos in conversations with OpenAI model 5.6 Sol, and Gemini, in Sept 2026.